LAB 3

Task 1: Adding 2 numbers:

```
add.sh
~/lab3

read -p "Enter num1: " num1
read -p "Enter num2: " num2
(( result=$num1 + $num2 ))
echo "Result is: $result"
```

```
student@student-OptiPlex-7000:~/lab3$ ./add.sh
Enter num1: 5
Enter num2: 6
Result is: 11
```

Task 2: scripting:

```
script.sh
~/lab3

#!/bin/bash

SRC_DIR=/home/student/lab3/d1
DST_DIR=/home/student/lab3/d2

if [ ! -d "$SRC_DIR" ];
then
	echo "Error: Source directory does not exist"
	exit 1
fi

if [ ! -d "$DST_DIR" ];
then
	mkdir -p "$DST_DIR"
fi

for file in "$SRC_DIR"/*;
do
	cp "$file" "$DST_DIR"
done
```

```
student@student-OptiPlex-7000:~/lab3$ mkdir d1
student@student-OptiPlex-7000:~/lab3$ mkdir d2
student@student-OptiPlex-7000:~/lab3$ cd d1
student@student-OptiPlex-7000:~/lab3/d1$ touch f1.txt f2.txt
student@student-OptiPlex-7000:~/lab3/d1$ cd ..
student@student-OptiPlex-7000:~/lab3$ touch script.sh
student@student-OptiPlex-7000:~/lab3$ pwd
/home/student/lab3
student@student-OptiPlex-7000:~/lab3$ chmod +x script.sh
student@student-OptiPlex-7000:~/lab3$ ./script.sh
student@student-OptiPlex-7000:~/lab3$
```

Task 3: 1) Multiplication:

multiply.sh
~/lab3

```bash
#!/bin/bash

(( result = $1 * $2 ))
echo "Multiplication result is: $result"
```

```
student@student-OptiPlex-7000:~/lab3$ ./multiply.sh 4 2
Multiplication result is: 8
student@student-OptiPlex-7000:~/lab3$
```

2)

count.sh
~/lab3

```bash
#!/bin/bash

read -p "Enter a sentence: " s
count=0

for (( i=0; i<${#s}; i++ ))
do
    ch=${s:$i:1}

    if [[ $ch == "a" || $ch == "e" || $ch == "i" || $ch == "o" || $ch == "u" || $ch == "A" || $ch == "E" || $ch == "I" || $ch == "O" || $ch == "U" ]]
    then
        count=$((count + 1))
    fi
done

echo "Total vowels are: $count"
```

```
student@student-OptiPlex-7000:~/lab3$ ./count.sh
Enter a sentence: I study in fast university
Total vowels are: 8
student@student-OptiPlex-7000:~/lab3$
```

3)

file.sh
~/lab3

```bash
#!/bin/bash

read -p "Enter file name: " name

if [[ -f $name ]]
then
        echo "$(date)" >> name.txt
        echo "Date and time added successfully"
else
        echo "File not found"
fi
```

```
student@student-OptiPlex-7000:~/lab3$ ./file.sh
Enter file name: sample.txt
Date and time added successfully
```

4)

status.sh
~/lab3

```bash
#!/bin/bash

if [ -e "$1" ];
then
	echo "Exist"
else
	echo "Does not exist"
	exit;
fi

if [ -f "$1" ];
then
	echo "Type: File"
else
	echo "Type: Directory"
fi

if [ -r "$1" ];
then
	echo "Readable"
else
	echo "Not readable"
fi

if [ -w "$1" ];
then
	echo "Writable"
else
	echo "Not writable"
fi

if [ -x "$1" ];
then
	if [ -d "$1" ];
	then
		echo "Executable directory"
	else
		echo "Executable file"
	fi
else
	echo "Not executable"
fi

if [ -o "$1" ];
then
	echo "Owned by current user"
else
	echo "Not owned by current user"
fi
```

```
simroz@simroz:~/lab3$ ./status.sh status.sh
Exist
Type: File
Readable
Writable
Executable file
Not owned by current user
simroz@simroz:~/lab3$
```

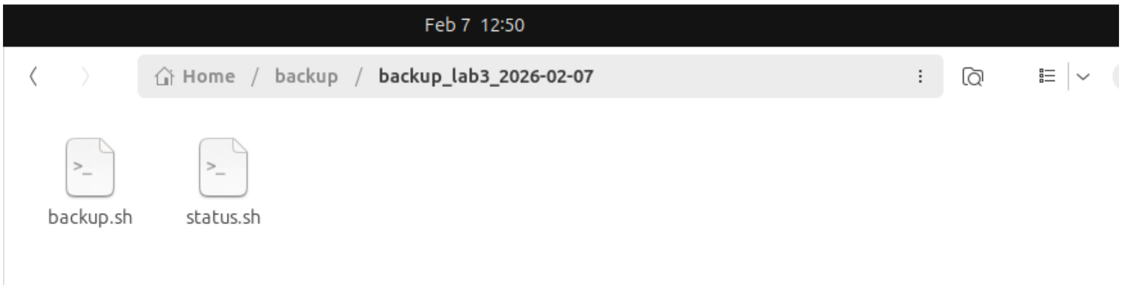
5)

Open backup.sh ~/lab3

```bash
#!/bin/bash

if [ $# -eq 0 ];
then
	echo "Error: Provide directory path"
	exit;
fi

if [ ! -d "$1" ];
then
	echo "Error: Directory not found"
	exit;
fi

current_date=$(date '+%Y-%m-%d' )
backup_name="backup_$(basename "$1")_$current_date"
backup_dir="$HOME/backup"

mkdir -p "$backup_dir"

file_count=$(find "$1" -type f | wc -l)

echo "Found $file_count files"

echo "creating backup: $backup_name"
cp -r "$1" "$backup_dir/$backup_name"
```

```bash
if [ $? -eq 0 ];
then
	echo "Backup completed"
	echo "Backup saved to: $backup_dir/$backup_name"
else
	echo "Backup failed"
fi
```

```
simroz@simroz:~/lab3$ ./backup.sh /home/simroz/lab3
Found 2 files
creating backup: backup_lab3_2026-02-07
Backup completed
Backup saved to: /home/simroz/backup/backup_lab3_2026-02-07
simroz@simroz:~/lab3$
```


Feb 7 12:50
Home / backup / backup_lab3_2026-02-07
backup.sh
status.sh